

Until today: Proved languages are regular:

- Constructed DFAs/NFAs.
- Proved closure properties.

...but what *can't* DFAs/NFAs do?

Today: We'll prove a few languages aren't regular.

Tricking a DFA

Question: How much information could a DFA with k states remember?

Finite. Can keep track of at most k things.

Idea: Finite memory $\rightarrow$ easy to trick.

Language: $A = \{a^n b^n \mid n \geq 0\}$

Example: Which of these strings are in this language?

- $ababababab$
- $aaaabbbb$
- $aaabbbb$
- $aaaab$
- ab

Intuition: This language requires memory to store a count of the a s.

Strategy:

- Assume $M = (Q, \{a, b\}, \delta, s, F)$ was a DFA for A .
- Goal: Prove that $L(M) \neq A$.

Non-sequitor: Is it possible that each of us was born in a different month?

Pigeonhole principle: Given n items and $m < n$ possible labels, at least two items must share a label.

- Items? People
- Labels? Birthday months

How does pigeonhole tell us something?

Assume M has k states.

Consider $a^k b^k$. M must accept $a^k b^k$ (or we're done)

Ex. $k = 7$.

- Items: a^i for $i \in \{0, \dots, k\}$
- Labels: $\hat{\delta}(s, a^i)$

Pigeonhole principle: There exists $i < j \leq k$, $p \in Q$ such that $\hat{\delta}(s, a^i) = \hat{\delta}(s, a^j) = p$.

Question: What is $\hat{\delta}(p, a^{j-i})$?

Informally: M loops on a^k .

Idea: Leave out this loop

Formally:

- $u = a^i$
- $v = a^{j-i}$ (the loop)
- $w =$ the rest of a^k
- $z = b^k$

Then:

- $\hat{\delta}(s, u) = p$
- $\hat{\delta}(p, v) = p$
- $\hat{\delta}(p, wz) = f \in F$

Question: What is $\hat{\delta}(s, uwz)$?

What about $\hat{\delta}(s, uv^2wz)$? $\hat{\delta}(s, uv^3wz)$?

Question: Which of the following are in A ?

- uwz
- $uvwz$
- uv^2wz
- uv^3wz

Our DFA accepts them all.

What did we do: M accepts strings in $A \Rightarrow M$ accepts other stuff, too.

Conclusion? $L(M) \neq A$. But M was arbitrary!

On Friday: Generalize this approach.

(optional) Pumping Lemma (informal statement). Assume you can do the following with some language L .

- Let k ($|Q|$) be given.
- Pick a test string $s = xyz \in L$, with a pigeonhole window y of length $> k$.
- Let a loop in y be given: $y = uvw$, with $|v| > 0$.
- Pick a number of times i such that $xuv^i wz \notin L$

Then L is not regular.

Closure Properties

Idea: Like reductions in 280 (can be used to prove something is tractable or intractable)

Let $B = \{x \mid \#a(x) = \#b(x)\}$

$A \subseteq B$. Does this imply that B is not regular?

Let $C = \{a^n b^m \mid n, m \geq 0\}$

Question: Write A in terms of B and C . $A = B \cap C$

Question: Come up with a DFA for C .

This means B can't be regular. Why?

If B was regular, then A would be regular, by closure under intersection.

A is not regular, so B cannot be regular.